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Liquid cooling device with a heat-resisting structure

Abstract

An integrated liquid cooling device with a heat-resisting structure includes an internal circulation type liquid cooling radiator, a liquid cooling head and a liquid pump. A liquid collection box of the internal circulation type liquid cooling radiator has an air heat insulation space inside, which divides the interior into a cold liquid chamber and a hot liquid chamber, so that the hot liquid flowing into the hot liquid chamber is prohibited from conducting heat to the cold liquid in the cold liquid, preventing the cold liquid from being abnormally heated and degrading the cooling effect. In addition, the two ends of the radiator pipes of the internal circulation type liquid cooling radiator form a respective trumpet pipe wall that is stuck in the liquid collection box, which can increase the strength of the connection in the liquid collection box.

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Background/Summary

BACKGROUND OF THE INVENTION

(a) Field of the Invention

(1) The invention relates to the field of heat dissipation for computers and servers, in particular to an integrated liquid cooling device with a heat-resisting structure, which can prevent heat transfer from hot liquid circulating inside the liquid cooling device to cold liquid.

(b) Description of the Prior Art

(2) The liquid cooling devices on today's computers and servers are usually composed of separate liquid cooling radiator, liquid cooling head, liquid pump and liquid pipes. The liquid pipes are connected between the liquid cooling radiator and the liquid cooling head, so that the coolant in the liquid cooling radiator and the liquid cooling head circulates. After the coolant absorbs heat on the liquid cooling head, it enters the liquid cooling radiator to dissipate heat, and the cooled cooling liquid flows back into the liquid cooling head. However, the liquid cooling radiator and the liquid cooling head of the above-mentioned liquid cooling device are arranged separately, and its structure is not compact, and it is relatively inconvenient to use. Moreover, there is no liquid pump function

on the liquid collecting tank of the liquid cooling radiator, so that the flow speed of the cooling liquid in the liquid cooling radiator is relatively slow, and the heat dissipation efficiency is low. Moreover, there is no separation in the liquid collection tank, so that the flow of the coolant in the liquid cooling radiator is relatively short, resulting in the inability of the coolant to effectively cool down and dissipate heat.

(3) Based on the above reasons, the inventor once proposed an integrated liquid cooling device (patent announcement U.S. Pat. No. 11,566,847B2) to solve the shortcomings of the conventional separated liquid cooling device. However, the liquid cooling radiator of the integrated liquid cooling device usually only uses a metal partition to separate the liquid collection box into a cold liquid chamber and a hot liquid chamber. When the liquid flowing through the liquid cooling head absorbs heat and turns into hot liquid and flows into the hot liquid chamber, it will conduct heat to the cold liquid in the adjacent cold liquid chamber through the metal partition, causing the cold liquid that is going to flow into the liquid cooling head to be heated first. Therefore, it affects the cooling effect after it flows into the liquid cooling head. Furthermore, the pipe and liquid collection box of the known liquid cooling radiator are welded and combined, and the inlet and outlet of the pipes do not take into account the fluidity of the liquid inflow and outflow, resulting in turbulent flow and flow velocity effects. In addition, the conventional liquid cooling radiator only relies on the liquid collection box connected to the two ends of the liquid pipes, and there is no other auxiliary design of the liquid collection box connecting the two ends, and its structural strength is easily affected by the wall strength of the liquid pipes. The thinner the pipe wall will lead to the more unstable structure of the overall liquid cooling radiator.

SUMMARY OF THE INVENTION

(4) The main object of the present invention is to provide an integrated liquid cooling device with a heat-resisting structure, which can prevent the hot liquid heat in the hot liquid chamber from being transferred to the cold liquid in the adjacent cold liquid chamber, prevent the cold liquid from absorbing heat and increasing the temperature abnormally, and improve the cooling efficiency of the liquid cooling device through the structural design of a first liquid collection box, a second liquid collection box and radiator pipes connected between the first liquid collection box and the second liquid collection box of an internal circulation type liquid cooling radiator and an air heat insulation space set in the first liquid collection box, the integrated liquid cooling device.

(5) The second object of the present invention is to provide an integrated liquid cooling device with a heat-resisting structure, wherein through the assembly structure design of the internal circulation type liquid cooling radiator, the pipe walls at both ends of the radiator pipes form a respective trumpet pipe wall that expands outward, and the trumpet pipe walls can hold the flat through holes of the first liquid collection box and the second liquid collection box to improve the structural strength of their combination and the smoothness of the working fluid flowing into and out of the radiator pipes.

(6) Another object of the present invention is to provide an integrated liquid cooling device with a heat-resisting structure, wherein there are locking lugs protruding from both ends of the fan mounting frames, and the locking lugs are respectively embedded in the concave holes of the first liquid collection box and the second liquid collection box to enhance the structural strength of the combination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a three-dimensional schematic diagram of a preferred embodiment of the present invention.

(2) FIG. 2 is a schematic diagram of the front view of the preferred embodiment of FIG. 1 of the

present invention.

(3) FIG. 3 is a cross-sectional schematic diagram of the preferred embodiment of FIG. 2 of the present invention.

(4) FIG. 4 is an exploded schematic view of the preferred embodiment of FIG. 1 of the present invention.

(5) FIG. 5 is a three-dimensional schematic diagram of the assembly structure of the pipe end of the present invention.

(6) FIG. 6 is a partial schematic diagram at C in FIG. 3 of the present invention.

(7) FIG. 7 is a schematic diagram of the internal rib structure of the second box body of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(8) The present invention is an integrated liquid cooling device with a heat-resisting structure, which is used to replace the traditional air cooling device installed on the processor inside the computer or server, and liquid-cool the processor to prevent the temperature from being too high. As shown in FIG. 1 and FIG. 2, its preferred embodiment comprises an internal circulation type liquid cooling radiator **10**, a liquid cooling head **20** and a liquid pump **30**.

(9) The internal circulation type liquid cooling radiator **10** preferably implements a first liquid collection box **11**, a second liquid collection box **12** and a plurality of first radiator pipes **13** and second radiator pipes **14**. The first liquid collection box **11** and the second liquid collection box **12** are hollow metal boxes respectively used for liquid collection, such as boxes made of aluminum alloy. The plurality of first radiator pipes **13** and second radiator pipes **14** are flat metal pipes whose two ends are respectively fixedly combined with the first liquid collection box **11** and the second liquid collection box **12**, and connected to the first liquid collection box **11** and the second liquid collection box **12**, so that the working liquid in the first liquid collection box **11** flows into the second liquid collection box **12** through the first radiator pipes **13**, and then flows into the first liquid collection box **11** through the second radiator pipes **14**, forming the closed internal circulation type liquid cooling radiator. In addition, the plurality of first radiator pipes **13** and second radiator pipes **14** are provided with a plurality of heat dissipation fins **15**, so that the working liquid can dissipate heat when flowing through the first radiator pipes **13** and second radiator pipes **14**. The liquid cooling head **20** is an existing item, such as the liquid cooling head shown in the patent announcement U.S. Pat. No. 11,566,847B2 or other liquid cooling heads that can be directly combined with the liquid cooling radiator. It is fixedly combined with the outside of the first liquid collection box **11**, and is used for cooling the surface of the processor in the computer or server. The liquid pump **30** is arranged in the second liquid collection box **12**, which has a known motor and impeller structure, and can be used to drive the circulating flow of working fluid in the internal circulation type liquid cooling radiator **10** to pass through the liquid cooling head **20** for cooling (heat absorption), and dissipate heat when circulating in the first radiator pipes **13** and second radiator pipes **14**.

(10) As shown in FIG. 3 and FIG. 4, an air heat insulation space **111** is set inside the above-mentioned first liquid collection box **11** of the present invention, and the air heat insulation space **111** partitions the interior of the first liquid collection box **11** into a cold liquid chamber **112** and a hydrothermal chamber **113**. The box wall of the cold liquid chamber **112** is provided with a cold liquid inlet **114** for the working liquid to flow from the cold liquid chamber **112** to the liquid cooling head **20**. The box wall of the hot liquid chamber **113** is provided with a hot liquid inlet **115** for returning the working liquid from the liquid cooling head **20** to the hot liquid chamber **113**. The air heat insulation space **111** is composed of a hot liquid partition **116** and a cold liquid partition **117** arranged inside the first liquid collection box **11** and separated by a distance. The periphery of the hot liquid partition **116** and the periphery of the cold liquid partition **117** are welded on the inner surface of the first liquid collection box **11**, and are located at the position between the first radiator pipes **13** and the second radiator pipes **14** (as shown in FIG. 3), so that the above-

mentioned air heat insulation space **111** is formed between the hot liquid partition **116** and the cold liquid partition **117**. Since the thermal conductivity of air is 0.025 W/m.Math.K, and the thermal conductivity of aluminum is 237 W/m.Math.K, the air heat insulation space **111** of the present invention can effectively prevent the heat transfer of the hot liquid flowing into the hot liquid chamber **113** to the cold liquid in the cold liquid chamber **112** next to it, preventing the cold liquid from being abnormally heated and ensuring the cooling efficiency of the cold liquid flowing into the liquid cooling head **20**.

(11) As shown in FIG. 3 and FIG. 4 again, the above-mentioned first liquid collection box **11** specifically comprises a first box body **118** and a first box cover **119**. The first end of the first box body **118** is a tapered closed end, the liquid cooling head **20** is fixed outside the tapered closed end, the second end of the first box body **118** opposite to its first end is an open end. The first box cover **119** covers the open end of the first box body **118**, and the first box cover **119** is provided with first flat through holes **1191** corresponding to the plurality of first radiator pipes **13** and second radiator pipes **14**. The periphery of the above hot liquid partition **116** and the periphery of the cold liquid partition **117** are in close contact with three inner surfaces of the first box body **118** and the inner surface of the first box cover **119**, thereby forming the above air heat insulation space **111**. The above-mentioned second liquid collection box **12** specifically comprises a second box body **121** and a second box cover **122**. The second box body **121** is a rectangular box, the first end of the second box body **121** is a closed end, and the second end of the second box body **121** opposite to the first end is an open end. The second box cover **122** covers the open end of the second box body **121**, and the second box cover **122** is provided with second flat through holes **1221** corresponding to the plurality of first radiator pipes **13** and second radiator pipes **14**.

(12) In order to improve the combined strength of the above-mentioned first radiator pipes **13** and second radiator pipes **14** in the above-mentioned first flat through holes **1191** and second flat through holes **1221**, and at the same time improve the smoothness of the liquid flowing in and out, taking the end of the first liquid collection box shown in FIG. 4, FIG. 5 and FIG. 6 as an example, in the present invention, the first ends of the plurality of first radiator pipes **13** and second radiator pipes **14** protrude from the outer wall of the first box cover **119** through the first flat through holes **1191** to the inner surface of the first box cover **119**, and make the pipe walls of the first ends of the plurality of first radiator pipes **13** and second radiator pipes **14** form a respective trumpet pipe wall **131**, **141** that expands outwards. The outer wall surfaces of the trumpet pipe walls **131** and **141** abut against the surroundings of the first flat through holes **1191**, so that they are not only stuck in the first flat through holes **1191**, but also improve the smoothness of the fluidity of the liquid. Similarly, when the trumpet pipe walls **132**, **142** the second ends of the first radiator pipes **13** and the second radiator pipes **14** of the present invention are combined with the second flat through holes **1221** of the second liquid collection box **12**, the same structure as above can also be adopted.

(13) As shown in FIG. 1 and FIG. 4 again, the preferred embodiment of the present invention further comprises two fan mounting frames **40**, and the two fan mounting frames **40** are combined on both sides of the internal circulation type liquid cooling radiator **10**. Each of the two ends of each fan mounting frame **40** respectively protrudes a locking lug **41** (as shown in FIG. 6), and the locking lugs **41** are respectively embedded in a respective concave hole **110**, **120** corresponding to the position of the first liquid collection box **11** and the second liquid collection box **12**. More specifically, each fan mounting frame **40** is integrally punched and formed from an aluminum alloy plate, and has a side plate **42** corresponding to the two sides of the internal circulation type liquid cooling radiator **10**, and at least one panel **43** connected to at least one side of the side plate **42** (As shown in FIG. 4, it is two panels **43**). The panel **43** is used to extend to one or both sides of the internal circulation type liquid cooling radiator **10**, and a fan fixing hole is arranged on the panel **43**. The two ends of the above-mentioned side plate **42** are respectively bent with an end plate **44** that fits on the outer walls of the first liquid collection box **11** and the second liquid collection box **12**. The free ends of the two end plates **44** are bent to form the above locking lugs **41**. Therefore,

they are possible to embed the concave holes **110**, **120** corresponding to the first liquid collection box **11** and the second liquid collection box **12**. In this way, the assembly structural strength of the fan mounting frames **40** can be enhanced, and the overall structural strength of the internal circulation type liquid cooling radiator **10** can also be enhanced.

(14) As shown in FIG. 3 and FIG. 7, the second box body **121** of the second liquid collection box **12** of the present invention can be integrally formed with ribs **123** protruding from the inner or outer surface of the box wall. The structural strength of the second box body **121** can be improved through the structure of the ribs **123**. Similarly, the above first box body **118** can also implement the same rib structure. Furthermore, the above-mentioned second box body **121** is provided with a liquid injection hole **124** and a hole plug **125** on the wall of the second box body **121**, which are convenient for injecting and replenishing the working liquid of the liquid cooling device.

(15) Although a particular embodiment of the invention has been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the invention. Accordingly, the invention is not to be limited except as by the appended claims.

Claims

1. An integrated liquid cooling device with a heat-resisting structure, comprising an internal circulation type liquid cooling radiator, a liquid cooling head and a liquid pump, wherein: said internal circulation type liquid cooling radiator comprises a first liquid collection box, a second liquid collection box, a plurality of first radiator pipes and a plurality of second radiator pipes; said first radiator pipes and said second radiator pipes being flat metal pipes with respective two ends thereof respectively combined with said first liquid collection box and said second liquid collection box and respectively connected to the inside of said first liquid collection box and the inside of said second liquid collection box; said first radiator pipes and said second radiator pipes being respectively externally provided with a plurality of heat dissipation fins; said first liquid collection box is internally provided with an air heat insulation space, said air heat insulation space dividing the interior of said first liquid collection box into a cold liquid chamber and a hot liquid chamber; said cold liquid chamber comprising a cold liquid inlet for a working liquid to flow from said cold liquid chamber to said liquid cooling head, said hot liquid chamber comprising a hot liquid inlet for said working liquid to flow back from said liquid cooling head to said hot liquid chamber, said air heat insulation space being composed of a hot liquid partition and a cold liquid partition arranged inside said first liquid collection box and separated by a distance, said hot liquid partition and said cold liquid partition being welded on the inner wall of said first liquid collection box, and being located between said first radiator pipe and said second radiator pipe, so that said air heat insulation space is formed between said hot liquid partition and said cold liquid partition; and said liquid cooling head is combined on the outside of said first liquid collection box, and said liquid pump is set in said second liquid collection box to drive said working liquid in said internal circulation type liquid cooling radiator to pass through said liquid cooling head for heat absorption and heat dissipation, wherein said first liquid collection box comprises a first box body and a first box cover, said first box body comprising a first end and an opposing second end, the first end of said first box body being a tapered closed end, the second end of said first box body being an open end, said first box cover covering the open end of said first box body, said first box cover being provided with a plurality of first flat through holes corresponding to said first radiator pipes and said second radiator pipes; said liquid cooling head is fixed on the outside of said tapered closed end of said first box body of said first liquid collection box; said hot liquid partition and said cold liquid partition are in close contact with three inner walls of said first box body and an inner wall of said first box cover.

2. The integrated liquid cooling device with a heat-resisting structure as claimed in claim 1,

wherein the first ends of said first radiator pipes and said second radiator pipes protrude from an outer wall of said first box cover to the inner wall of said first box cover after passing through the respective said first flat through holes, the pipe walls of the first ends of said first radiator pipes and said second radiator pipes respectively forming a trumpet pipe wall that expands outward, said trumpet pipe wall having an outer wall surface thereof abutting against the periphery of one respective said first flat through hole.

3. The integrated liquid cooling device with a heat-resisting structure as claimed in claim 1, wherein said second liquid collection box comprises a second box body and a second box cover, said second box body being a rectangular box body, said second box body comprising a first end and an opposing second end, the first end of said second box body being a closed end, the second end of said second box body being an open end, said second box cover covering the open end of said second box body, said second box cover being provided with a plurality of second flat through holes corresponding to said first radiator pipes and said second radiator pipes.

4. The integrated liquid cooling device with a heat-resisting structure as claimed in claim 3, wherein the second ends of said first radiator pipes and said second radiator pipes protrude from an outer wall of said second box cover through said second flat through holes to an inner wall of said second box cover, the pipe walls of the second ends of said first radiator pipes and said second radiator pipes forming a respective trumpet pipe wall that expands outward, the trumpet pipe walls of said first radiator pipes and said second radiator pipes having a respective outer wall surface thereof abutting against the periphery of one respective said second flat through hole.

5. The integrated liquid cooling device with a heat-resisting structure as claimed in claim 3, wherein said second box body is integrally formed with a plurality of ribs protruding from inner or outer box wall thereof.

6. The integrated liquid cooling device with a heat-resisting structure as claimed in claim 3, wherein said second box body is provided with a liquid injection hole and a hole plug on a wall of said second box body.

7. The integrated liquid cooling device with a heat-resisting structure as claimed in claim 1, further comprising two fan mounting frames respectively combined on both sides of said internal circulation type liquid cooling radiator, each said fan mounting frame comprising two locking lugs respectively protruding from both ends thereof, and said locking lugs being respectively embedded in one respective concave hole of said first liquid collection box and one respective concave hole of said second liquid collection box.

8. The integrated liquid cooling device with a heat-resisting structure as claimed in claim 7, wherein each said fan mounting frame comprises a side plate corresponding to two sides of said internal circulation type liquid cooling radiator, at least one panel connected to at least one side of said side plate and extending to one side of said internal circulation type liquid cooling radiator and a fan fixing hole arranged on said panel, said side plate having two ends thereof respectively bent with an end plate that fits on the outer walls of said first liquid collection box and said second liquid collection box, said end plates having free ends thereof bent to form the respective said locking lugs.
